Economy Simulator

Mathematical Models and Algorithms

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Abstract

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1 Introduction

1.1 Project Overview

1.2 Mathematical Objectives

2 Crafting and Production Model

2.1 Mathematical Definitions

2.2 Construction of the Consumption Matrix

The purpose of the consumption matrix C is to represent the quantities of items required to execute each crafting recipe.

Let n denote the total number of items supported by the game, and let m denote the total number of crafting recipes. The consumption matrix is therefore defined as $C \in \mathbb{Z}_{\geq 0}^{n \times m}$.

Each row i, where $1 \leq i \leq n$, represents an item, while each column j, where $1 \leq j \leq m$, represents a crafting recipe.

Thus, the entry C_{ij} specifies the quantity of the ith item consumed when executing the jth recipe once. If the recipe does not require that item, then $C_{ij} = 0$.

2.3 Construction of the Production Matrix

Similar to the consumption matrix, the production matrix B's purpose is to represent the quantities of items produced after executing each crafting recipe. B has the same dimensions as C, because the rows and columns represent the same items and recipes as C, except the entries may be different because B catalogues the number of item outputs whereas C catalogues the number of item inputs.

Therefore, $B \in \mathbb{Z}_{\geq 0}^{n \times m}$, where the entry B_{ij} specifies the quantity of the ith item produced when executing the jth recipe once. If the recipe does not produce that item, then $B_{ij} = 0$.

2.4 Mixed-Integer Linear Programming: Limitations of the Current Model

The current model is limited because it is given a daily production capacity and minimum quotas for each product. This means that after fulfilling all quotas for each day, each corporation will manufacture the most profitable item until its daily production capacity is filled.